

Summer 2026 Combinatorial Mathematics and Graph Theory with JavaScript MATH:510

INSTRUCTOR: Dr. Stefan Forcey

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OFFICE HOURS: By appointment!

Text and Coverage: (Three books, for recommended reading only)

Eloquent Javascript by Haverbeke (free, see website).

Winning Ways for your Mathematical Plays by Berlekamp, Conway, and Guy.

Introductory Combinatorics by Brualdi.

Website for schedule, lectures, homework problems and announcements:

https://sforcey.github.io/sf34/class_home/comb/combgamesjs.htm

GRADING POLICY:

- There are multiple homework quizzes and 2 tests. Most of the credit for each will be based on completion. All incorrect answers will be marked, but the score will be at least an 80% for each completed test or quiz question. Zero points will be earned for any missing items, so make sure to turn in all the work!

- The quiz/homework average will be calculated by dropping a total of 15 quiz points which means that I'll calculate your percentage by first adding up to 15 points back on to your score, limited by the maximum number of hw/quiz points possible. This will have the effect of making a 100% quiz average possible despite some missed.

- There will be 2 tests during the semester over the material from lectures.

- 1000 points possible as follows:

800 pts: Homework, quizzes (80%)

200 pts: 2 Tests at 100 pts each. (20%)

(points earned will be the same fraction of the total in each graded category as the average fractional score earned on items in that category)

900 pts. guarantees an A

800 pts. guarantees a B

700 pts. guarantees a C

600 pts. guarantees a D

(+,- at my discretion)

Course Outline:

Day 1, July 13: 1.1 JavaScript

Day 2, July 14: 1.2 Games

Day 3, July 15: 1.3 Sets

Day 4, July 16: 2.1 Combinatorics 1

Day 5, July 17: 2.2 Combinatorics 2

Day 6, July 20: 2.3 Combinatorics 3

Day 7, July 21: 2.3 3.1 Combinatorics 4

Day 8, July 22: 2.3 3.2 Combinatorics 5

Day 9, July 23: 2.3 3.3 Combinatorics 6

Day 10, July 24: 2.3 3.4 Combinatorics 7

Day 11, July 27: 2.3 4.1 Combinatorics 8

Day 12, July 28: 2.3 4.2 Combinatorics 9

Day 13, July 29: 2.3 4.3 Combinatorics 10

Day 14, July 30: 2.3 4.4 Combinatorics 11

Test 1: July 31

Day 15, July 31: 2.3 5.1 Combinatorics 12

Day 16, Aug 3: 2.3 5.2 Combinatorics 13

Day 17, Aug 4: 2.3 5.3 Combinatorics 14

Day 18, Aug 5: 2.3 6.1 Graph theory 1

Day 19, Aug 6: 2.3 6.2 Graph theory 2

Test 2 Aug 7.

Homework and tests may not be copied, but collaboration and research are allowed.

For information on "WHAT STUDENTS NEED TO KNOW," go to [What Students Need To Know : The University of Akron, Ohio \(uakron.edu\)](https://uakron.edu/what-students-need-to-know) (see the list of items below).

- The Student Code of Conduct and academic misconduct
- Statement about the ethical use of ChatGPT and other AI tools
- Inclusive Excellence
- Title IX
- Sexual harassment and sexual violence
- Students with disabilities
- Religious accommodations for students
- ZipAssist